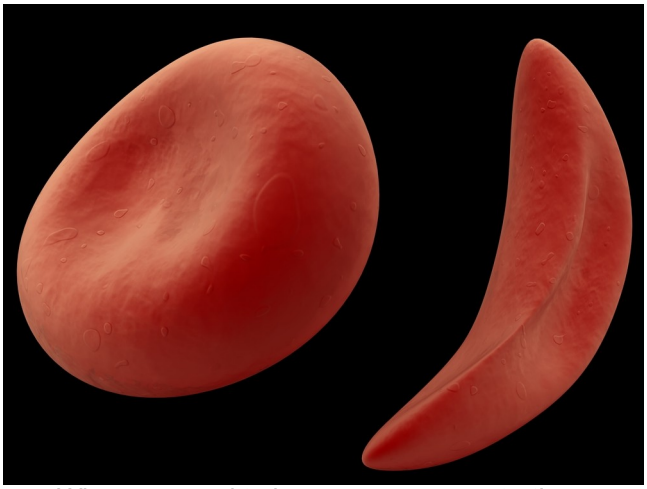
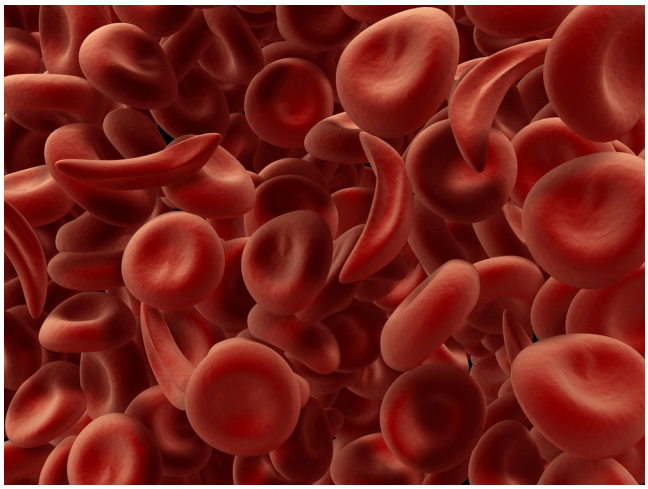
Sickle cell disease results from a mutation on the 11th chromosome that causes a valine molecule to be substituted for glutamic acid in the Beta-globin protein. Normal adult hemoglobin (hemoglobin A) consists of two Alpha-globin proteins and two Beta-globin proteins. In patients with sickle cell disease, the two Beta-globin proteins are malformed. This abnormal hemoglobin is referred to as 'hemoglobin S'.

Patients who are heterozygous for the trait are referred to as having 'sickle cell trait' and their symptoms are not as severe as those who are homozygous, but must still be addressed with caution under anesthesia and in any situation where hypoxia is a possibility.



When exposed to low oxygen concentrations, hemoglobin S precipitates and causes the red blood cell to deform into a sickle shape.



The sickled cells accumulate in capillaries, causing a mechanical obstruction to blood flow that can manifest as pain, ischemia, and infarction of distal tissues.

Of all the possible sequelae of sickle cell disease, the most significant threat is acute chest syndrome. With a mortality rate as high as 20%, acute chest syndrome begins as dyspnea, chest pain, and hypoxemia and can progress to adult respiratory distress syndrome. Factors that place the sickle cell patient at risk for acute chest syndrome include pulmonary infection, fat embolism, asthma exacerbations, and perioperative respiratory dysfunction.

The risk of anesthesia is higher in patients who have experienced this syndrome in the past, who are older than 30 years, exhibit hypoxia at rest, or have a history of perioperative complications due to sickle cell disease.

Methods to reduce complications of sickle cell disease include: adequate oxygenation, prevention of vascular stasis, normothermia, adequate hydration, and prevention of increased oxygen consumption (shivering or hyperthermia). More involved procedures usually require transfusion of red blood cells to maintain a hematocrit between 30% and 35%. Both general and regional anesthesia are acceptable in most cases, but a Bier block should be avoided in patients homozygous for the disease.

Barash PG, Cullen BF, Stoelting RK. Clinical Anesthesia. 6th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2009: 634-636.

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